SCIENCE AND MEDICINE IN FOOTBALL
2022, VOL. 6, NO. 3, 355–362
https://doi.org/10.1080/24733938.2021.1965201

The effectiveness of 3D multiple object tracking training on decision-making in soccer

Sebastian Harenberg[a], Zachary McCarver[b], Justin Worley[c], Dennis Murr[d], Justine Vosloo[e], Rumit Singh Kakar[f], Rob McCaffrey[g], Kim Dorsch[g] and Oliver Höner[d]

[a]Department of Human Kinetics, St. Francis Xavier University, Antigonish, NS, Canada; [b]School of Sport and Exercise Science, University of Northern Colorado, Greeley, CO, United States; [c]Department of Kinesiology, Michigan State University, East Lansing, MI, United States; [d]Department of Sport Psychology and Research Methods, Eberhard Karls University, Tübingen, Germany; [e]Department of Exercise Sciences and Athletic Training, Ithaca College, Ithaca, NY, United States; [f]Department of Human Movement Science, Oakland University, Rochester, MI, United States; [g]Faculty of Kinesiology, University of Regina, Regina, SK, Canada

ABSTRACT

Background: Soccer requires athletes to make quick decisions in dynamic environments. Several off-court technology-based interventions have been developed to train these perceptual cognitive skills. However, the evidence for training transfer using technologies to athletic performance has been sparse. Previous research found 3-dimensional multiple object tracking (3D MOT) training to cause a significant increase in quality of passing decision-making. Limitations to the research warrant further investigation of this association.
Purpose: To re-examine the effectiveness of 3D MOT on training decision-making.
Methods: Thirty-one NCAA Division III soccer players (female n = 16) were randomized to 3D MOT training or a control task. The experimental group received 10 training sessions over a span of 4 weeks.
Results: The manipulation check indicated a significant training effect in 3D MOT performance for the intervention but not the control group ($F_{(1,29)} = 21.46$, $p < .001$, $n_p^2 = .43$). Non-significant changes with small effect sizes ($n_p^2 = .01$-$.03$) in decision-making and measures of near-transfer were found.
Conclusion: The findings challenge the association between 3D MOT training and increased quality of decision-making in soccer.

ARTICLE HISTORY
Accepted 22 July 2021

KEYWORDS
Perceptual cognitive skills; multiple object tracking; technology; collegiate athletes; decision making

Introduction

Soccer athletes execute movements under immense time constraints in highly dynamic environments. The ability to anticipate the flow of a game (e.g., opponent's or ball movement) and make appropriate decisions based on this information are vital for success (Williams and Jackson 2019; Williams et al. 2020). Accordingly, Mann et al. (2007) defined perceptual-cognitive skills as 'the ability to identify and acquire environmental information for the integration of existing knowledge such that appropriate responses can be selected and executed' (p. 457). These skills include several cognitive processes (e.g., anticipation, decision-making, visual search, pattern recognition) that soccer athletes apply in sport-specific situations. Several models (Cañal-Bruland & Mann, 2015; Raab 2002; Williams 2009) exist that explain the interrelation of perceptual cognitive skills for decision-making and anticipation under pressure. The current evidence indicates that experts are generally superior in the execution of such skills compared to novice athletes (Mann et al. 2007).

The advancement of technology permits the display of dynamic visual scenes in immersive contexts, such as 3D visualization and virtual reality, using electronic devices (e.g., computers, head-mounted displays). As such, athletes can be trained cognitively off the field in addition to regular practices. Cognitive training in athletes has become a popular area for research and business. Several products are commercially available, ranging from LED Light equipment to Virtual Reality applications. While the diversity of products to train perceptual cognitive skills has increased drastically, evidence of the effectiveness of such modalities is still developing (Zentgraf et al. 2017).

A central question that remains largely unclear is which type of technology-assisted perceptual cognitive training may provide the best transfer effect to a specific task or sport. Theories explaining the underlying mechanisms of transfer between technology-assisted perceptual cognitive training and real-life performance in sport are sparse (Müller and Rosalie 2019). The appropriate design of perceptual cognitive training tasks with a targeted transfer effect to a cognitive or motor skill in mind remains unclear. To provide a better understanding, Hadlow et al. (2018) proposed the Modified Perceptual Training Framework, providing a theoretical foundation for the evaluation of different technologies. In particular, the authors argue that the transfer of perceptual-cognitive training depends on three factors:

The first factor is the perceptual function that is trained. In other words, the complexity of the visual-cognitive skill similar

CONTACT Sebastian Harenberg sharenbe@stfx.ca St. Francis Xavier University, Department of Human Kinetics, 4130 University Ave, Antigonish, Canada

to the targeted task in real-life. For example, a VR football task may not accurately display some important contextual cues that would normally play a role in making decisions on the field. The second factor is the visual similarity of the presented stimulus. That is, how similar is the visual scene presented in the technology is to real life. As an example, players and objects in a virtual environment may look and move very differently compared to real-life. The third factor is the response to the stimulus. This factor captures the representativeness of the athlete's interaction with the virtual world. For example, in some VR environments, the player can walk 'freely' in the virtual world by using designed passive treadmills whereas in other virtual environments movement is impossible or requires the use of handheld controllers. Hadlow et al. (2018) argue that technologies high in sport-specific similarity on these three factors should have larger transfer effects to sport performance compared to technologies that are low in similarity.

Because of an absence in theory in the design of many commercially available cognitive training devices, it is unsurprising that the existing preliminary evidence on their effectiveness for cognitive training is mixed. The current evidence has been summarized in two systematic reviews (Zentgraf et al. 2017; Harris et al. 2018). Both reviews showed that most studies suffered from important limitations, including different frequencies of training, lack of power, and absence of active control groups. While the reviewed studies did not indicate a uniform transfer effect to performance, a few technologies showed more promise than others.

One training modality that was favorably mentioned by both reviews is three-dimensional multiple object tracking (3D MOT). The technology targets different domains of attention (e.g., sustained, divided) and processing speed, which are claimed to be important for cognitive performance in soccer (Faubert and Sidebottom 2012). In the seminal study in a team sport athlete population, Faubert (2013) demonstrated that professional athletes had superior abilities to train on the 3D MOT software compared to semi-professional and amateur-level athletes. The study provided first evidence that high-performing athletes may be superior in rapidly learning complex and dynamic visual scenes. Since its release, the training and transfer effects of 3D MOT have been tested in several sports (Magine et al. 2014; Faubert and Barthes 2018; Romeas et al. 2019), and other performance domains (e.g., laparoscopic surgery; Harenberg et al. 2016).

For soccer, 3D MOT training may be particularly applicable because athletes need to track multiple objects (e.g., ball, opponents, teammates) frequently. Romeas et al. (2016) examined the effectiveness of 3D MOT training on decision-making. Twenty-three participants from a Canadian university soccer team were assigned into an experimental (n = 9), an active (n = 7) group, and a passive control (n = 7) group. The experimental group was trained on 3D MOT for 10 times (for approximately 25 minutes) over a five-week span. The active control group watched 10 20-minute 3-dimensional highlight videos of the 2010 FIFA World Cup followed by a quiz. The passive group did not receive any interventions. Despite the different treatments for active and passive control groups, the authors decided to combine the groups for analysis purposes. An adapted decision-making coding instrument that ranks the quality of passing, shooting, and dribbling decisions in a series of 5 vs. 5 soccer games (Gabbett et al. 2008) was used as the main dependent variable. The 3D MOT group improved passing decision-making accuracy, while the combined control group maintained their score from pre- to post-test. Differences in dribbling and shooting decision-making were non-significant. The authors interpreted the findings as support for transfer of 3D MOT to soccer decision-making performance.

While Romeas et al. (2016) findings may have important implications for cognitive training in soccer athletes, it is important to note that the study was not free of limitations. For example, Vater et al. (2021) criticized the assessment of decision-making used in the study. While the video-taped on-field assessment method has been used in previous research (Gabbett et al. 2008), the reliability and validity of the assessment has yet to be demonstrated. In addition, the authors only used a single assessor, which further limits the validity of the findings. Changes in near-transfer effects (e.g., by employing other measures of attention or processing speed) were not assessed. Second, a theoretical interpretation of the evidence for far-transfer was absent. It remains unclear why the transfer only occurred in passing decision-making and not in the other tested domains (shooting and dribbling), which may equally require the tracking of multiple objects (Vater et al. 2021). Third, Romeas et al. (2016) recruited only a small sample size ($n = 23$, < 10 athletes per group; Zentgraf et al. 2017; Vater et al. 2021), which was not sufficiently justified (e.g., via a power analysis). Lastly, the combination of the active and passive control groups, most likely to enhance statistical power, may be questionable because no placebo treatment (i.e., 3D video) was administered to the passive control group. Taken together, Vater et al. (2021) stated that 'recommending this "training effect" to soccer coaches seems inadequate with the explained limitations' (p. 15).

In light of the limitations of Romeas et al. (2016) study, further investigation in the effectiveness of 3D MOT training to enhance decision-making in a soccer context with a larger sample size, more objective assessments, and a control task for all participants in the control group may be warranted (Vater et al. 2021). This is the purpose of the present study. Based on the theoretical assumptions of 3D MOT (Faubert and Sidebottom 2012) and existing evidence (Romeas et al. 2016; Fleddermann et al. 2019), it was hypothesized that 3D MOT training will lead to positive effects in near (i.e., measures of attention) and far-transfer (i.e., decision-making).

Methods

Participants

Participants were recruited from soccer teams at a single NCAA Division III institution. A priori power analysis using G*Power 3.1 revealed a required sample size of 28 participants to detect a previously reported effect size, as published by Romeas and colleagues (i.e., η_p^2 = .16), at

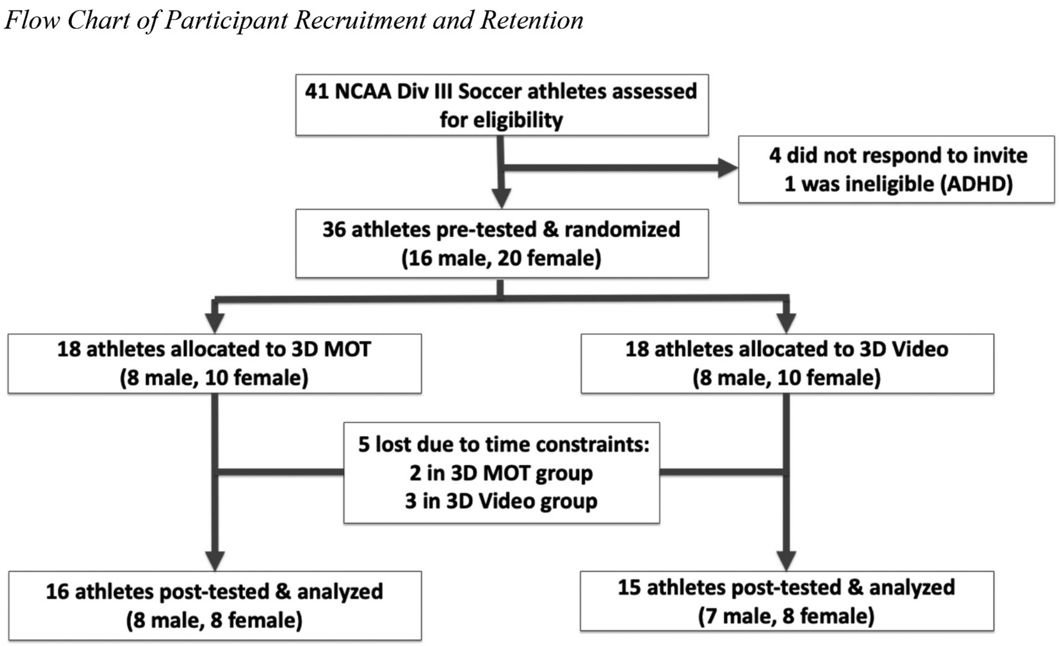

Figure 1. Flow Chart of Participant Recruitment and Retention

Table 1. Descriptives of the Sample Demographics and Dependent Variables.

Variable	All (N = 31)	Experimental (n = 16)	Control (n = 15)	p, ES
Age	19.13 (.92)	19.31 (1.14)	18.93 (.59)	.40
Sex	Male = 15 Female = 16	Male = 8 (50%) Female = 8 (50%)	Male = 7 (46.67%) Female = 8 (53.33%)	.85
Experience Soccer (in years)	13.97 (2.04)	13.50 (1.97)	14.47 (2.07)	.14
Experience Collegiate Soccer (in years)	1.68 (.79)	1.75 (.86)	1.60 (.86)	.68
3D MOT Pre (speed threshold)	1.50 (.39)	1.52 (.40)	1.47 (.38)	.75
3D MOT Post (speed threshold)	1.84 (.58)	2.22 (.51)	1.44 (.34)	<.001, g = 1.72
Stroop Pre (correct responses)	48.45 (10.73)	49.88 (11.12)	46.93 (10.46)	.25
Stroop Post (correct responses)	54.65 (13.12)	55.13 (14.33)	54.13 (12.19)	.11
Trails B Pre (in seconds)	52.81 (12.22)	53.33 (12.91)	52.26 (11.88)	.81
Trails B Post (in seconds)	43.58 (10.59)	45.19 (11.97)	41.87 (8.98)	.31
Decision Making Pre (% Accuracy)	60.82 (10.20)	59.11 (9.34)	62.64 (11.07)	.35
Decision Making Post (% Accuracy)	63.44 (11.51)	62.76 (11.63)	64.17 (11.74)	.74

Continuous variables are presented in Means and Standard Deviations, Categorical variables are presented in counts and percentages.

a power of .80. Athletes were included if they were at least 18 years of age. Exclusion criteria were a diagnosis of ADHD, a concussion in the past 6 months, and severe visual impairment (e.g., unable to read font size 16). Forty-one athletes attended the information session. Out of the 41 athletes, 36 decided to take part in the study. Due to time constraints, 31 (86.1%) participants (female n = 16, male n = 15) completed the study. A flow chart of the recruitment and retention is presented in Figure 1. The demographics of the sample are summarized in Table 1.

Procedure

Ethical approval was obtained at the participating institution. The coaches of the male and female soccer teams provided permission to invite their athletes. Participants completed a consent form and were scheduled for a pre-test appointment. The athletes were randomized into the experimental (3D MOT) or active control group (3D Videos) via statistical software. The stimuli for the experimental and control group were consistent with those described in Romeas et al. (2016). The athletes completed the pre-test and were scheduled for 10 training sessions over a four-week span. At the end of the training sessions, athletes were scheduled for a post-test. All tests and training sessions were conducted by trained research assistants.

Treatment stimuli

3D MOT. The core module of the 3D MOT software (Neurotracker, Cognisens, Montreal, QC, Canada) was used for the intervention group. The task involves tracking several targets on a screen simultaneously, and is described in depth elsewhere (Faubert and Sidebottom 2012; Faubert 2013; Romeas et al. 2019). The participants in the intervention group were placed on a chair in front of a 55-inch 3D TV (Sony Model XBR55X930D) at a distance of 56 inches and were wearing active 3D glasses. The intervention included 10 trainings (consisting of three sessions with 20-trial repetitions of 3D MOT), which were each approximately 25–30 minutes in length.

3D Videos. The 3D videos were game replays of the group and final stages of the 2010 FIFA World Cup. Similar to the intervention group, the control group were placed on a chair in front of a 55-inch 3D TV (Sony Model XBR55X930D) while wearing 3D glasses. The participants watched scenes from different games for approximately 25 minutes for each session. For the final six sessions, participants re-watched the scenes and were interviewed afterwards about specific aspects of the games.

Measurement

Similar to Romeas et al. (2016), 3D MOT changes from pre- and post-test between the experimental and control group were analyzed as a manipulation check. As the Neutrotracker is supposed to train various domains of attention, extensively validated psychological tests of selective and divided attention as well as processing speed were chosen to measure the near-transfer of the 3D MOT training. Similar forms of near-transfer tests have been used in previous research (Fleddermann et al. 2019). As Romeas et al. (2016) claimed a far transfer to passing decision-making, a video-based test battery of passing situations was chosen.

Manipulation check

Each participant (intervention and control) completed one 3D MOT session consisting of three repetitions of 20 trials (Neurotracker – core module) at pre- and post-test. The program calculates the individual's speed threshold (i.e., a standardized score based on centimeters per second of the average ball movement on correct trials) of each repetition of 20 trials. The average of the three repetitions was calculated and used as the outcome measure. Due to the training, the experimental group should significantly increase their 3D MOT performance, while the performance of the control group should not.

Near-transfer

Stroop. The Stroop Test is included in the Delis-Kaplan Executive Function System (D-KEFS; Delis et al. 2001) and was used to assess selective attention of the participants. The test requires participants to look at a series of colored words, which are presented in incongruent colors. The participant was required to state the font color of the word font and not the color the word might state as fast as possible without errors. The recorded outcome for this study was the number of correct responses in a 60-second trial. Each participant completed a 10-second familiarization trial. An app version of the Stroop Test on a 10-inch tablet was used.

Trails B. The Trails Making Test – Version B was used as an assessment of processing speed, divided attention, and executive function (Tombaugh 2004). The test is part of the Halstead-Reitan Battery (Reitan and Wolfson 1985) and requires the participant to connect a sequence of numbers and letters (i.e., 1,A,2,B,3,C ...) as quickly as possible without errors. The outcome was recorded in total time in seconds. Each participant completed a brief familiarization trial before the test. The Trails B was administered in form of an app on a 10-inch tablet.

Far-Transfer

Decision-Making. For the assessment of decision-making, a video-based diagnostic battery with a soccer-specific motor response was used (Murr et al. 2021). The generation and validation of the diagnostic battery is described elsewhere (Murr et al. 2021). While viewing the videos (four different categories including 12 video scenes, respectively), the participant was instructed to be a player in ball possession. For each round, a dynamic scene with up to three teammates (in green) and three opponents (in red) unfolded in different locations on the pitch (i.e., four different video scene categories). The player was instructed to be on the team in green and play the ball to the player with the most space or shoot to the goal (i.e., in the striker category). The passing options developed to the left, center, or right of the participant's point of view. An auditory signal during the black screen indicated the start of a new video. Exactly six seconds later, a frozen image of a scene appeared. This image permitted the participant to get oriented in the video. The image was frozen for 1.5 seconds. Afterwards, the clip was played (each clip was between 5–6 seconds in length). Each situation had one player that clearly generated more space than the other options, and hence presented the best passing option. Once the situation was over, a black screen appeared.

In the laboratory, the videos were presented to the participants via a projector on a 254 cm-diagonal screen. The participants were placed 5 m away from the screen in a central position, with a ball on their feet. Three pop-up nets were placed centrally and to the left and right under the screen (see Figure 2). The player was instructed to play the ball to the left, center, or right, corresponding with the correct response identified in the video, as quickly as possible. As a familiarization, the participant was shown three trial videos in each category (which were not part of the test battery) before the 12 testing video scenes were demonstrated.

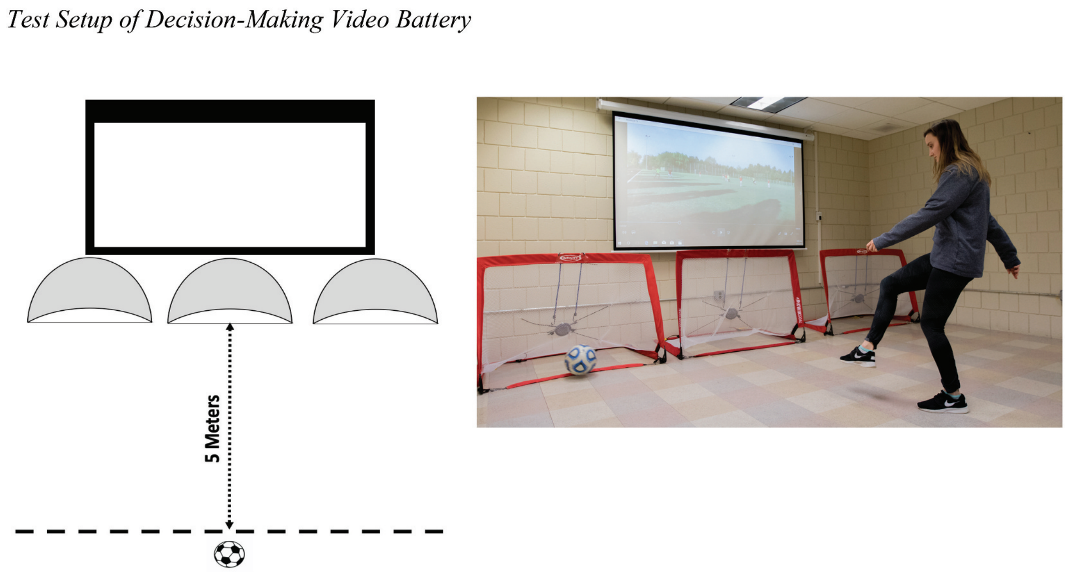

Figure 2. Test Setup of Decision-Making Video Battery Note: Distance in picture was reduced for demonstration

Table 2. Post-Hoc Comparisons from Pre- to Post-Test within Groups

	Control			Experimental		
Variable	Pre Test M (SD)	Post Test M (SD)	*p*, d	Pre Test M (SD)	Post Test M (SD)	*p*, d
3D MOT	1.47 (.38)	1.44 (.34)	1.00, .09	1.52 (.40)	2.22 (.51)	<.001, 1.35
Stroop	46.93 (10.46)	54.13 (12.19)	<.001, 1.47	49.88 (11.12)	55.13 (14.33)	.004, .86
Trails B	52.26 (11.88)	41.87 (8.98)	.024, .75	53.33 (12.91)	45.19 (11.97)	.068, .69
Decision Making	.63(.11)	.64 (.12)	1.00, .13	.59 (.09)	.63 (.12)	1.00, .29

Note: Holm correction p used for post-hoc tests, d = Cohen's d effect size

For the present study, the choices were recorded as 1 (correct decision) or 0 (incorrect decision) for each video scene. The outcome was the combined percentage of all correct choices across all 48 testing video scenes. Athletes were permitted to make a decision at least 1 second after the conclusion of the video or their trial would be marked as an error. If the athlete did not hit the target with the ball, it was marked as an error.

Statistical analysis

The characteristics of the experimental and control group at pre-test were compared with chi-square analysis for categorical data and independent t-tests for continuous data. All analyses with effect sizes are reported in Table 1. To compare the improvements in the dependent variables (i.e., 3D MOT, Stroop, Trails B, Decision-Making) between the groups over time, four mixed-models analyses of variance (ANOVA) with a between-group factor (i.e., group) and a repeated factor (i.e., time) were calculated. Eta squared effect sizes were calculated for all ANOVAs. Post-hoc tests were calculated using Holm's correction and Cohen's d effect sizes.

Results

Participant characteristics

Participants in the experimental and control group did not differ significantly in the demographic variables as well as the dependent variables at baseline (see Table 1).

Manipulation check

The analysis revealed that the experimental group but not the control group improved their 3D MOT speed threshold (Interaction effect: $F_{(1,29)} = 21.46$, $p < .001$, $\eta_p^2 = .43$; main effect for time: $F_{(1,29)} = 16.73$, $p < .001$, $\eta_p^2 = .37$; main effect for group: $F_{(1,29)} = 10.72$, $p = .003$, $\eta_p^2 = .27$), supporting the internal validity of the current intervention. Post-hoc analyses showed that the experimental group improved their 3D MOT performance by 46.1% ($p < .001$, d = 1.35) while the control group's performance only changed marginally (–2.1%, $p = 1.00$, d = –.09) from pre- to post-test. Post-hoc analyses are summarized in Table 2. All individual and group changes are displayed in Figure 3.

Changes in Individual and Group Scores across Time for all Tests

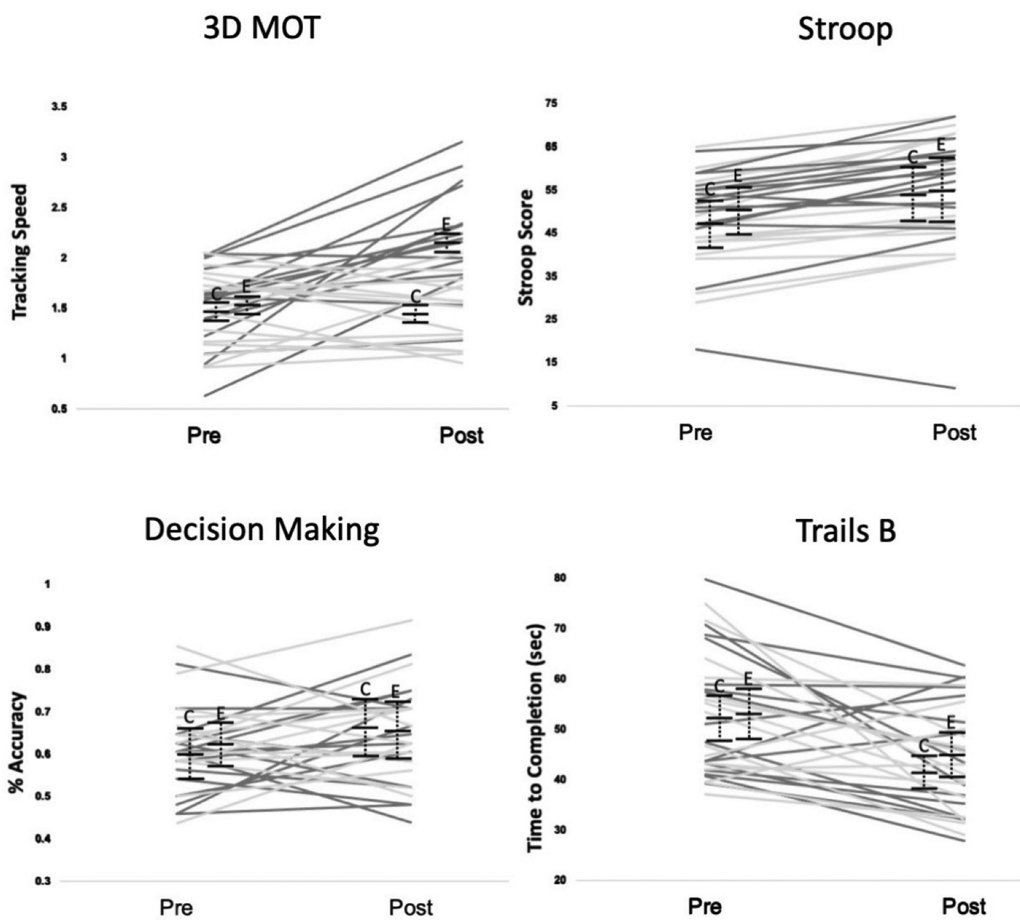

Figure 3. Changes in Individual and Group Scores across Time for all Tests Note: light grey = control; dark grey = experimental; black = means with 95% confidence intervals

Near-transfer

Stroop Test. The ANOVA revealed a significant main effect for time ($F_{(1,29)}$ = 39.04, p < .001, η_p^2 = .57). However, the main effect for group ($F_{(1,29)}$ = .21, p = .65, η_p^2 = .01) and the interaction effect were non-significant ($F_{(1,29)}$ = .26, p = .34, η_p^2 = .03). Taken together, this indicates that both groups improved their correct responses on the Stroop test at a similar rate over time. Accordingly, post-hoc tests revealed significant changes from pre- to post-test for the experimental (p = .004, d = .86) and the control group (p < .001, d = 1.47) with medium to large effect sizes.

Trails B. The ANOVA revealed a significant main effect for time ($F_{(1,29)}$ = 16.09, p < .001, η_p^2 = .36). However, the main effect for group ($F_{(1,29)}$ = .40, p = .53, η_p^2 = .01) and the interaction effect were non-significant ($F_{(1,29)}$ = .24, p = .63, η_p^2 = .01). The findings indicate that both groups improved in the speed of completion of the task in a similar fashion over time. Post-hoc tests revealed a significant improvement from pre- to post-test on the Trails B task in the control group (p = .02, d = .75), while the improvement in the experimental approached significance (p = .07, d = .69). Both improvements showed medium effect sizes.

Far-transfer

Decision-making. The ANOVA revealed a non-significant main effect for time ($F_{(1,29)}$ = 1.44, p = .24, η_p^2 = .05) and for group ($F_{(1,29)}$ = .56, p = .46, η_p^2 = .02). The interaction effect was non-significant as well ($F_{(1,29)}$ = .24, p = .63, η_p^2 = .01). This indicates that both groups did not improve on the decision-making task. Accordingly, the post-hoc tests for the experimental (p = 1.00, d = .29) and control group (p = 1.00, d = .13) from pre- to post-test were non-significant with small effect sizes.

Discussion

The purpose of the present study was to reexamine the impact of 3D MOT training on decision-making in soccer athletes. The present study utilized a similar interventional protocol of previous research (Romeas et al. 2016) and added some methodological improvements (i.e., larger sample size based on power analysis, near-transfer assessment, validated test of decision-making in soccer). While the manipulation check confirmed the enhancement of 3D MOT performance due to the intervention, the hypothesis that the 3D MOT training would lead to improvements in near- and far-transfer was not confirmed.

The experimental group experienced a significant gain in 3D MOT performance (46.1%), while the control group did not (−2.1%). While there was a significant improvement over time on the near-transfer measures, both groups improved at the same rate, indicating that the improvement could be due to repetition/training rather than the intervention. The lack of a near-transfer of 3D MOT training is in line with recent evidence. For example, Harris et al. (2020) also found no evidence of near transfer after a 12-week 3D MOT training in 84 undergraduate students. The present findings also stand in contrast to other studies. For example, Scharfen and Memmert (2021) found a small near-transfer effect after a 10-week 3D MOT training protocol in a sample of 29 elite youth soccer athletes. Fleddermann et al. (2019) found larger near-transfer effects in processing speed and sustained attention among a sample of 22 volleyball athletes. These finding could not be confirmed in the present study.

Moreover, the present study examined the far transfer effect of the 3D MOT training to decision-making. Both groups only improved marginally (experimental group: 3.65% and control group: 1.52%) and non-significantly on the decision-making task. The findings question a potential transfer of the 3D MOT training to decision-making in soccer athletes, which is consistent with existing research in sport. For example, two studies showed no far-transfer effects of 3D MOT training to decision-making performance in soccer (Scharfen and Memmert 2021) and volleyball (Fleddermann et al. 2019). However, the lack of a far-transfer effect in the present study contradicts other research evidence in sport. Most notably, the transfer effect of 3D MOT to passing decision-making detected in Romeas et al. (2016) study could not be replicated in the present study.

The lack of far-transfer effects of 3D MOT may be attributable to the specificity of the training task. That is, 3D MOT may resemble some aspects of dynamic situations in soccer, yet, may not replicate the entire complexity of a sport-specific situation. Romeas et al. (2019) stated that 3D MOT 'lacks specificity and does not include the combination of perception and action that is typically present in the sport environment' (p. 2). Accordingly, Hadlow and colleagues' MPTF suggests that the transfer of 3D MOT should slightly exceed those of touch-boards (e.g., D2, Fitlights), but are inferior to more immersive approaches (e.g., virtual reality). As such, VR-supported perceptual cognitive training may be more advantageous to train decision-making in soccer as the technology permits the display of scenes that are more sport-specific and realistic in visualization and task response. Future research in this area is warranted.

The findings of the present study should be interpreted in light of its limitations. While the study employed a randomized-controlled design, the lack of blinding of participants may have affected the results. Unfortunately, blinding was not possible because athletes were recruited from a male and female soccer team from the same institution. Because of the frequent interactions between teammates, there may have been information exchanged on the training conditions. The recruitment from a single NCAA Division III institution further limits the generalizability of the present study to other contexts. While Romeas et al. (2016) recruited from a similar population (i.e., Canadian university soccer athletes), higher-level athletes may respond differently to the 3D MOT training (Faubert 2013). In addition, the use of a passive control group would improve the design of the present study because it may reveal any change in task performance without a placebo treatment. Future research with larger sample sizes to test treatments in multiple experimental and control groups may be warranted. Similar to Romeas et al. (2016) the present study employed a rather short intervention (10 training session over 4 weeks), which are more feasible in a busy university sport season. Research examining longitudinal effects of 3D MOT is presently absent but is an area worth exploring in the future.

A last limitation revolves around the assessment of transfer used in this study. For near-transfer, standardized and well-validated tests of attention were used. Because the experimental and control group improved significantly on both tests, a learning effect is very likely, inhibiting the assessment of the contribution of 3D MOT on the tested cognitive domains. While a practice trial for both tests were provided to all participants at each test time, a possible avenue to prevent the presence of a learning effect may be the provision of additional practice trials before the actual measurement.

While standardization for the assessment of far transfer provides an important advantage for internal validity, it should be noted that the external validity of the test, in other words the real-life representativeness, may be jeopardized. The scenes of the chosen test only mimicked a handful of sport-specific situations that a player may encounter during a soccer match, but not the entire complexity of a soccer competition. This issue is common in transfer testing in the perceptual cognitive training literature. Williams (2019) argued that a truly representative design does not only capture potential links between perception and action but rather captures the full demands of competition with the highest degrees of specificity. For example, tests could include competition-specific stressors, such as crowd noise or a stadium atmosphere. Virtual reality applications may be used to introduce some of these stressors during tests. Because of the requirements in resources to create such tests, such as VR programming experience, it was deemed impossible for the present study. Nonetheless, future studies in perceptual cognitive training may utilize more representative and specific designs in transfer testing.

The current study adds to the mixed results of the transfer of perceptual cognitive training, and particularly 3D MOT, to enhance decision-making in soccer athletes (Zentgraf et al. 2017; Harris et al. 2018). The results did not confirm an association of 3D MOT and decision-making in soccer. Researchers, sport psychology practitioners, coaches, and athletes should be aware that 3D MOT and other modalities of perceptual training may, if at all, only marginally improve perceptual cognitive skills and cognitive performance in sport. This interpretation aligns with the review of Vater et al. (2021), which suggests that the promotion of the Neurotracker as a training tool for athletes is premature as its near- and far-transfer effects are 'not there or not very solid' (p. 22) However, the effectiveness of 3D MOT might be improvable by combining it with other tasks. For example, 3D MOT could be performed with background videos of sport scenes or actual skill execution (Romeas et al. 2019). Whether 3D MOT dual-task approaches could be effective to improve decision-making in soccer may be the aim of future research endeavors.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

No funding is associated with this study.

ORCID

Sebastian Harenberg http://orcid.org/0000-0003-0748-1823

References

Cañal-Bruland R, Mann DL. 2015. Time to broaden the scope of research on anticipatory behavior: a case for the role of probabilistic information. Front Psychol. 6:1–3. doi:10.3389/fpsyg.2015.01518

Delis DC, Kaplan E, Kramer JH. 2001. Delis Kaplan Executive Function System: examiner's manual. San Antonio (TX): Psychological Corporation.

Faubert J. 2013. Professional athletes have extraordinary skills for rapidly learning complex and neutral dynamic visual scenes. Sci Rep. 3(1):22–24. doi:10.1038/srep01154.

Faubert J, Barthes S. 2018. Prior perceptual-cognitive training builds mental resistance during acute physical fatigue in professional rugby athletes. BioRxiv. 505313. doi:10.1101/505313

Faubert J, Sidebottom L. 2012. Perceptual-cognitive training of athletes. J Clin Sport Psychol. 6(1):85–102. doi:10.1123/jcsp.6.1.85.

Fleddermann M, Heppe H, Zentgraf K. 2019. Off-court generic perceptual-cognitive training in elite volleyball athletes: task-specific effects and levels of transfer. Front Psychol. 10:1599. doi:10.3389/fpsyg.2019.01599

Gabbett TJ, Carius J, Mulvey M. 2008. Does improved decision-making ability reduce the physiological demands of game-based activities in field sport athletes? J Strength Condition Res. 22(6):2027–2035. doi:10.1519/JSC.0b013e3181887f34.

Hadlow SM, Panchuk D, Mann DL, Portus MR, Abernethy B. 2018. Modified perceptual training in sport: a new classification framework. J Sci Med Sport. 21(9):950–958. doi:10.1016/j.jsams.2018.01.011.

Harenberg S, McCaffrey R, Butz M, Post D, Howlett J, Dorsch KD, Lyster K. 2016. Can Multiple Object Tracking Predict Laparoscopic Surgical Skills? J Surg Educ. 73(3):386–390. doi:10.1016/j.jsurg.2015.11.013.

Harris DJ, Wilson MR, Smith SJR, Meder N, Vine SJ. 2020. Testing the effects of 3D Multiple Object Tracking training on near, mid and far transfer. Front Psychol. 11:1–7. doi:10.3389/fpsyg.2020.00196

Harris DJ, Wilson MR, Vine SJ. 2018. A systematic review of commercial cognitive training devices: implications for use in sport. Front Psychol. 9:709. doi:10.3389/fpsyg.2018.00709

Magine GT, Hoffman JR, Wells AJ, Gonzalez AM, Rogowski JP, Townsend JR, Stout JR. 2014. Visual tracking speed is related to basketball-specific measures of performance in NBA players. J Strength Condition Res. 28 (9):2406–2412. doi:10.1519/JSC.0000000000000550.

Mann DTY, Williams AM, Ward P, Janelle CM. 2007. Perceptual-cognitive expertise in sport: a meta-analysis. J Sport Exerc Psychol. 29(4):457–478. doi:10.1123/jsep.29.4.457.

Müller S, Rosalie S. 2019. Transfer of expert visual-perceptual-motor skill in sport. In: Williams AM, Jackson RC, editors. Anticipation and Decision Making in Sport. Routledge; p. 375–394. New York, USA.

Murr D, Larkin P, Höner O. 2021. Decision-making skills of high-performance youth soccer players. German J Exerc Sport Res. 51(1):102–111. doi:10.1007/s12662-020-00687-2.

Raab M. 2002. T-ECHO: model of decision making to explain behaviour in experiments and simulations under time pressure. Psychol Sport Exerc. 3 (2):151–171. doi:10.1016/S1469-0292(01)00014-0.

Reitan RM, Wolfson D. 1985. The Halstead–Reitan Neuropsycholgical Test Battery: therapy and clinical interpretation. Tucson (AZ): Neuropsychological Press.

Romeas T, Chaumillon R, Labbe D, Faubert J. 2019. Combining 3D-MOT with sport decision-making for perceptual-cognitive training in virtual reality. Percept Mot Skills. 1–27. doi:10.1177/0031512519860286

Romeas T, Guldner A, Faubert J. 2016. 3D-Multiple Object Tracking training task improves passing decision-making accuracy in soccer players. Psychol Sport Exerc. 22:1–9. doi:10.1016/j.psychsport.2015.06.002

Scharfen HE, Memmert D. 2021. Cognitive training in elite soccer players: evidence of narrow, but not broad transfer to visual and executive function. German J Exerc Sport Res. 51(2):135–145. doi:10.1007/s12662-020-00699-y.

Tombaugh TN. 2004. Trail making test A and B: normative data stratified by age and education. Archives Clin Neuropsychol. 19(2):203–214. doi:10.1016/S0887-6177(03)00039-8.

Vater C, Gray R, Holocombe A. 2021. A critical systematic review of the Neurotracker perceptual-cognitive training tool. Pschyo Bulleting Rev. 2021. doi:10.3758/s13423-021-01892-2

Williams AM. 2009. Perceiving the intentions of others: how do skilled performers make anticipation judgments? Prog Brain Res. 174:73–83. doi:10.1016/S0079-6123(09)01307-7

Williams AM. 2019. Perceptual-cognitive expertise and simulation-based training in sport. In: Hodges NA, Williams AM, editors. Skill Acquisition in Sport. Taylor & Francis; p. 237–254.New York, USA: Routledge.

Williams AM, Casanova F, Toledo I. 2020. Anticipation. In: Zeigler-Hill V, Shackelford TK, editors. Encyclopedia of Personality and Individual Differences. 1st ed. Berlin: Springer; doi:10.1007/978-3-319-28099-8.

Williams AM, Jackson RC. 2019. Anticipation in sport: fifty years on, what have we learned and what research still needs to be undertaken? Psychol Sport Exerc. 42:16–24. doi:10.1016/j.psychsport.2018.11.014

Zentgraf K, Heppe H, Fleddermann M-T. 2017. Training in interaktiven Sportarten. German J Exerc Sport Res. 47(1):2–14. doi:10.1007/s12662-017-0441-8.